CS3300 A: Introduction to Software Engineering

Lecture 14: Software Testing; Black-Box Testing

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Slides adapted from Alessandro Orso

Verification and Validation…

AI can verify well?



Requirements Engineering



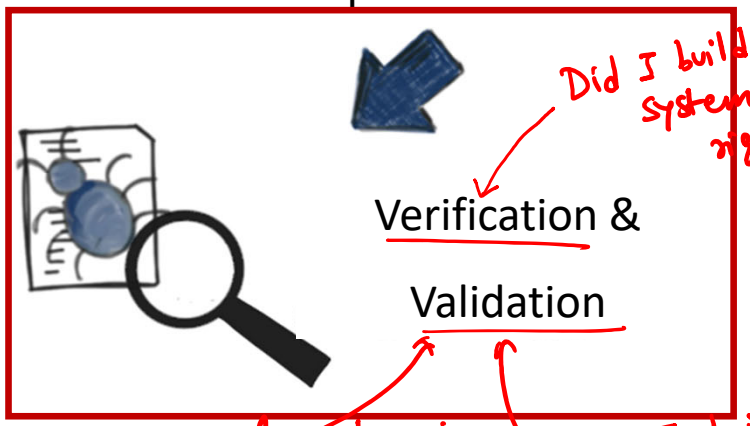
Design



Implementation



Maintenance



Why Verification Matters in the era of AI ?

High level system design & Verification skills

Code	As code generation becomes easier, engineers must be able to:
Define	Define expected software behavior.
Design	Design meaningful tests.
Identify	Identify incorrect assumptions and missing edge cases.
Evaluate	Evaluate AI-generated code and tests.
Assess	Assess whether the implementation satisfies requirements.

Pre-AI: Testing pays off in ~3 weeks

Post-AI: Testing pays off in ~3 days

-Titus Winters (senior principal scientist at Adobe focusing on developer experience)
Based on survey from Industry leaders, October, 2026

The Changing Role of Software Engineers

01

AI tools are increasingly used for implementation, debugging, and code comprehension.

02

Google has reportedly piloted AI-assisted coding and code-comprehension interviews.

03

Evaluating AI-assisted work requires technical understanding, debugging, and verification skills.

Google replaced one traditional coding round with a Code Comprehension round in its May 2026 pilot. Instead of solving a LeetCode-style algorithm from scratch, you receive a 200-500 line unfamiliar codebase in a CoderPad environment, and you have 60 minutes to read it, find bugs, discuss design decisions, and suggest improvements, all while using Google's own Gemini as an AI assistant. If you have been practicing only algorithmic problem-solving, this round will catch you off guard. Here is exactly what the round tests, why it exists, and how to build the skills it requires before your interview day.

https://www.linkedin.com/posts/ericcheng26_google-is-shifting-away-from-leetcode-style-share-7515118002262966272-mQoo/

Software is Buggy!

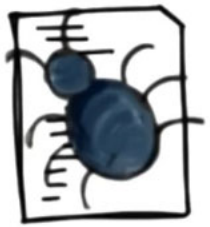
- Cost of bugs: $ 60 B/year
- On average, 1-5 errors per 1K LOC
- Windows 10
  - 50M LOC
  - 63,000 known bugs at the time of release
  - 1.25 per 1,000 lines

- For mass market software 100% correct SW development is infeasible, but
- We must verify the SW as much as possible

Failure, Fault, Error



Failure: Observable incorrect behavior of a program. Conceptually related to the behavior of the program, rather than its code.



Fault (bug): Related to the code. Necessary (not sufficient!) condition for the occurrence of a failure.



Error: Cause of a fault. Usually a human error (conceptual, typo, etc.)

Failure, Fault, Error: Example



```
1.    double doubleValue(int param) {
2.        double result;
3.        result = (double) param * param;
4.        return(result);
5.    }
```

param = 3
result = 9

+ * 2

Failure ⇒ Incorrect Behavior

A call to double(3) returns 9. What is this?	The result 9 is a failure- it is an observable behavior
Where is the fault?	Line 3
What is the error that caused the fault?	N/A. Maybe typo, erroneous copy paste, or conceptual. Only the developer knows.

human error or machine error

Approaches to Verification ≈ Testing

- **Testing** (dynamic verification): exercising software to try and generate failures

- **Static analysis**: identify (specific) problems statically, that is, considering all possible executions

- **Inspections/reviews/walkthroughs**: systematic group review of program text to detect faults

- **Formal verification** (proof of correctness): proving that the program implements the program specification

Testing → Dynamic Verification

exercise our software

Most difficult part: Choosing the input



Input Domain D | Software | Output Domain O

Test Case: {i ∈ D, o ∈ O}

Test Suite: A set of Test Cases

Test Case = { i, o }

Static Verification



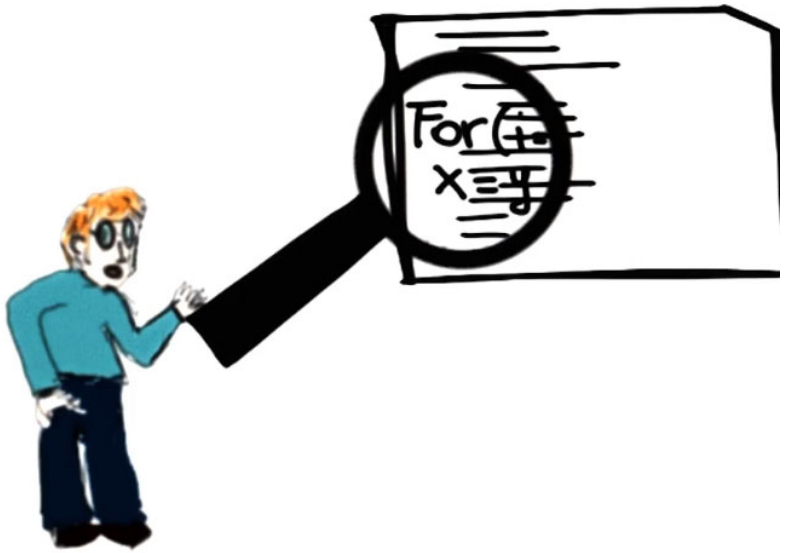
Input Domain D

Software

Output Domain O

Considers all possible inputs
(execution/behaviors)

Inspections/Reviews/Walkthroughs



Human intensive activity
Manual
Group activity
Inspect defects in the artifacts by identifying faults

Formal Proof (Of correctness)



Program

Specification

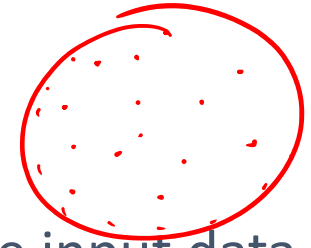
Given a formal specification, checks that the code corresponds to such specification

Sophisticated mathematical analysis

Comparison among the 4 techniques

	 PROS	 CONS
Testing	No False Positives	Highly Incomplete
Static Verification	Considers all program behaviors, Complete	False Positives, Expensive
Inspections	Systematic, Thorough	Informal, Subjective
Formal Proofs of Correctness	Strong Guarantees	Complex, Expensive to build/prove a mathematical basis

What is Testing?



Testing == To execute a program with a sample of the input data

- Dynamic technique: program must be executed
- Optimistic approximation:
  - The program under test is exercised with a (very small) subset of all the possible input data
  - We **assume** that the behavior with any other input is consistent with the behavior shown for the selected subset of input data

Successful Tests



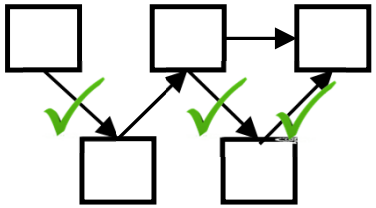
-Goodenough and Gerhart (1985). “Towards a Theory of Test data selection”. *IEEE Transactions of Software Engineering*, Jan 1985

Testing Granularity Levels

Unit Testing

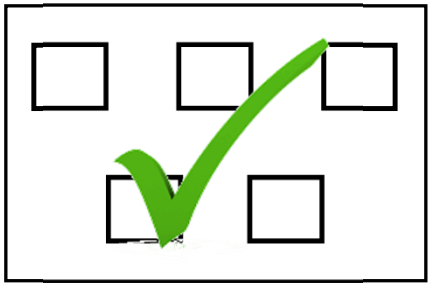


Integration



Big Bang

System



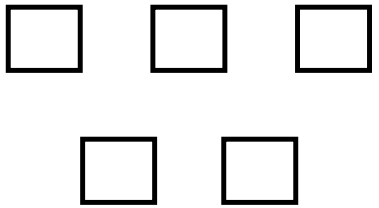
Functional/Non-functional

System

Acceptance Testing

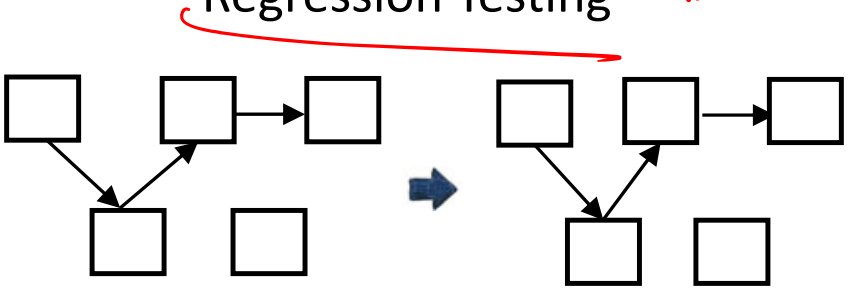


Customer



Most expensive & most imp.

Regression Testing



Testing Stages

✗ Def. Not. → Implement → Start Testing ✗

✓ Okay → Implement & Test → Together

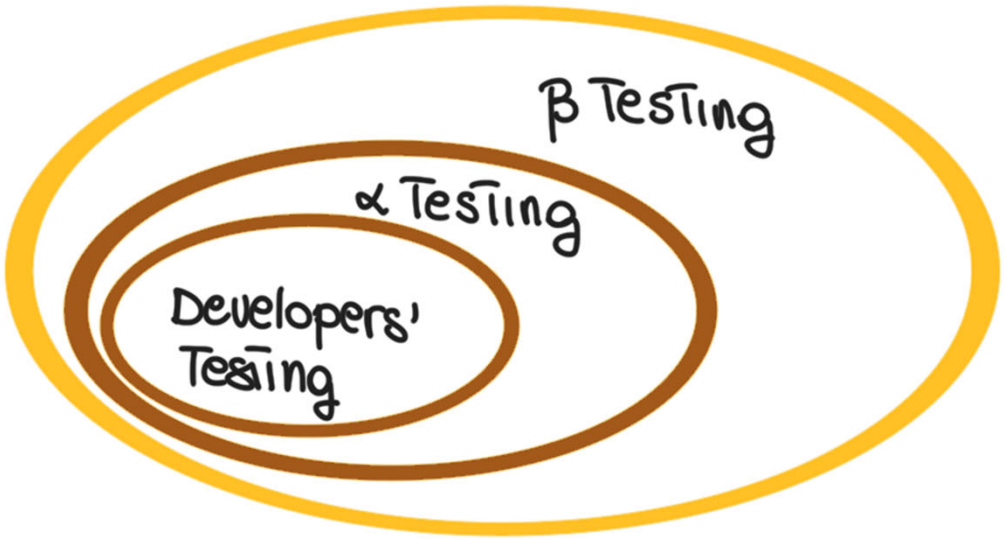
More popular → TDD → Test & then implement



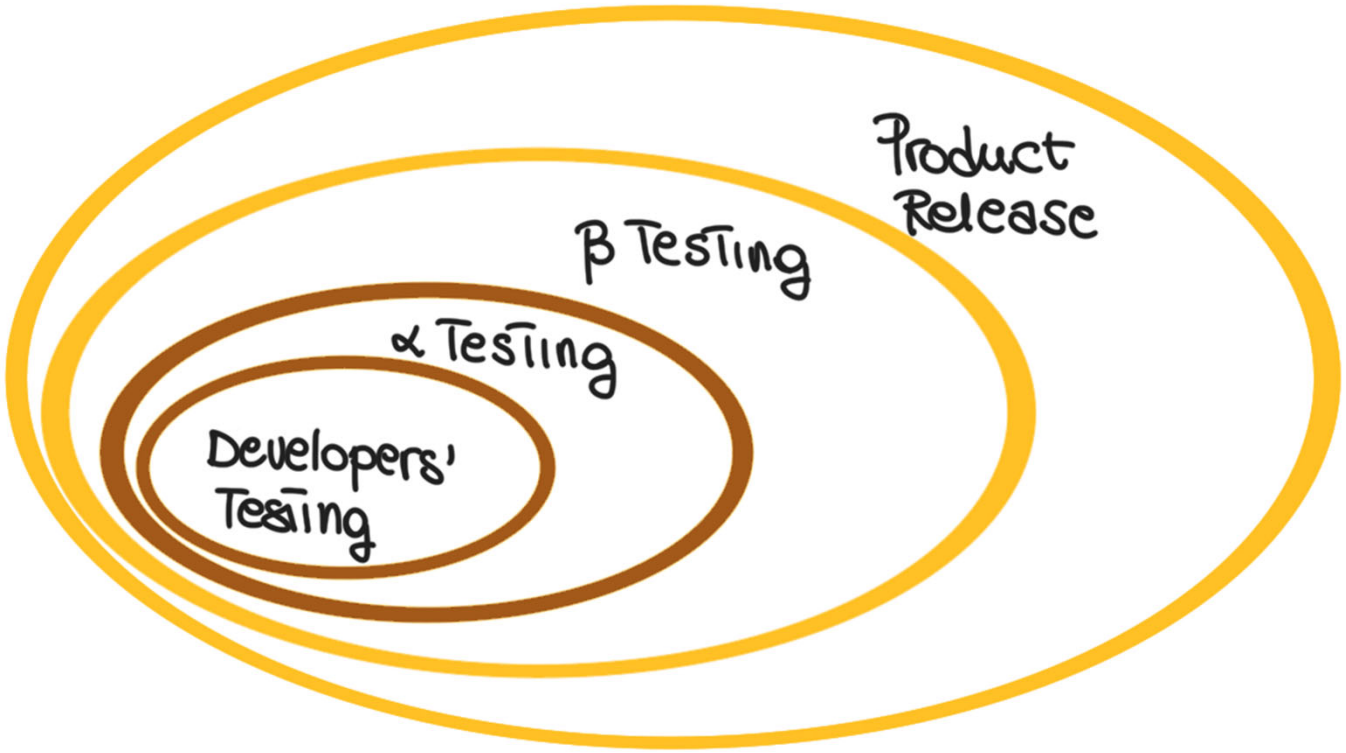
Testing Stages



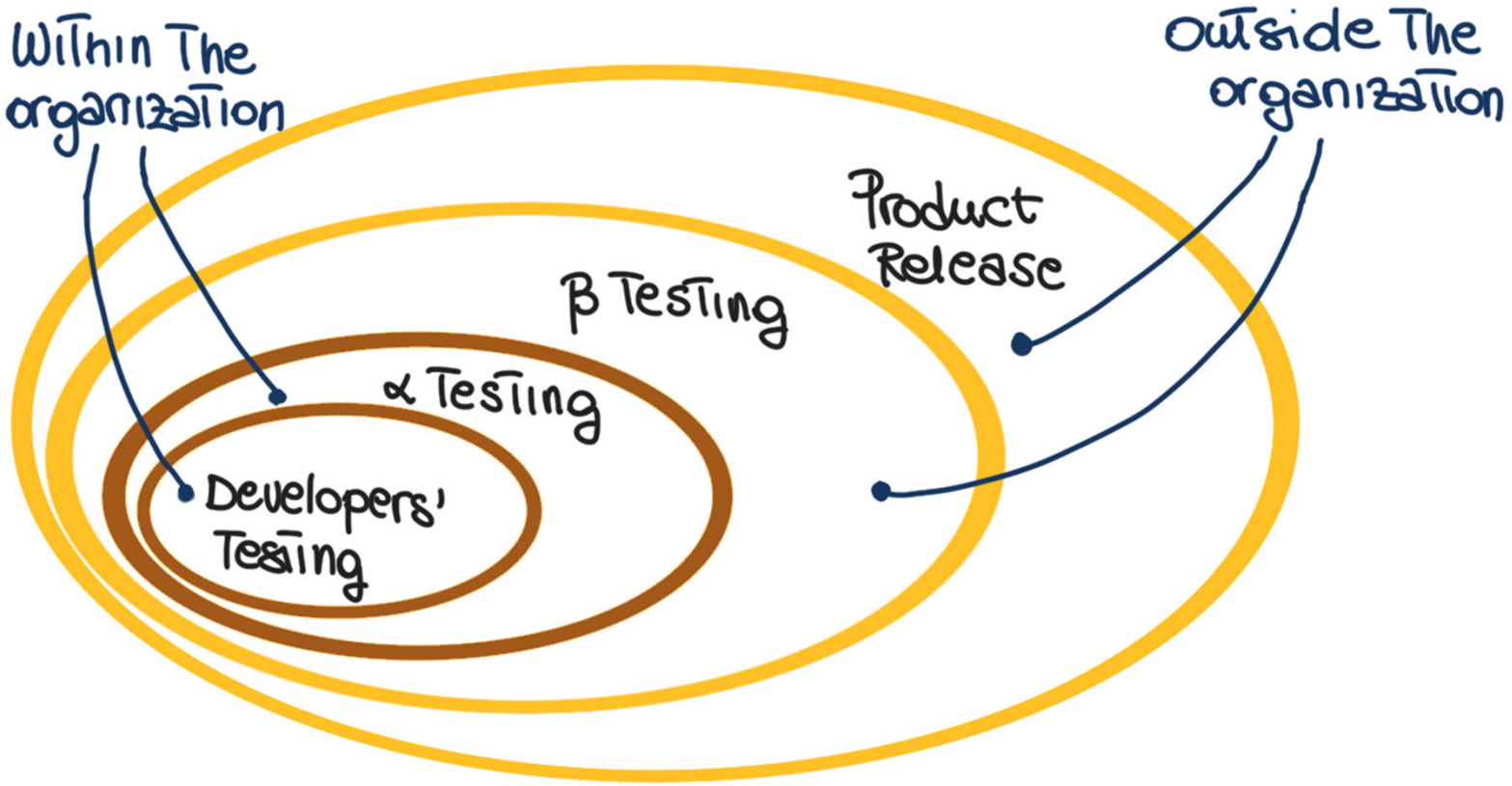
Testing Stages



Testing Stages



Testing Stages



Testing Techniques

There are several techniques

- Different processes
- Different artifacts
- Different approaches

There are no perfect techniques

- Testing is a best-effort activity

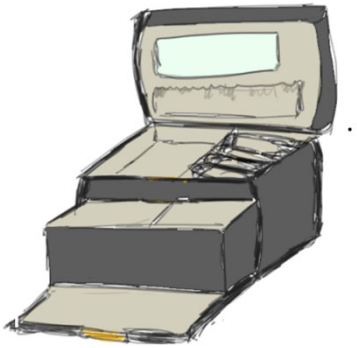
There is no best technique

- Different contexts
- Complementary strengths and weaknesses
- Trade-offs

Testing Techniques



code



BLACK BOX TESTING	WHITE BOX TESTING
Tests externally observable behavior	Tests based on internal code structure
Derived from specifications and requirements	Derived from statements, branches, conditions, and paths
Implementation knowledge not required	Requires implementation knowledge
Examples: equivalence partitions, boundary values	Examples: statement and branch coverage

Black-Box vs White box: Same Program, Different Tests

Specification: Display an integer byte count; values of 1024 or more should be displayed in KB.

Black-box

Tests derived from the specification.

- Positive
- Negative
- Zero
- Threshold values, if identified

```
void printNumBytes(int param) {
    if (param < 1024)
        printf("%d", param);
    else
        printf("%d KB", param / 124);
}
```

Black-Box vs White box: Same Program, Different Tests

Specification: Display an integer byte count; values of 1024 or more should be displayed in KB.

White-box

Tests derived from the code.

- param < 1024
- param >= 1024

Exercise both branches

```
void printNumBytes(int param) {
  if (param < 1024)
    printf("%d", param);
  else
    printf("%d KB", param / 124);
}
```

Which tests are likely to expose the incorrect divisor (124 instead of 1024)?

Black Box Testing - From Specifications to Test Cases

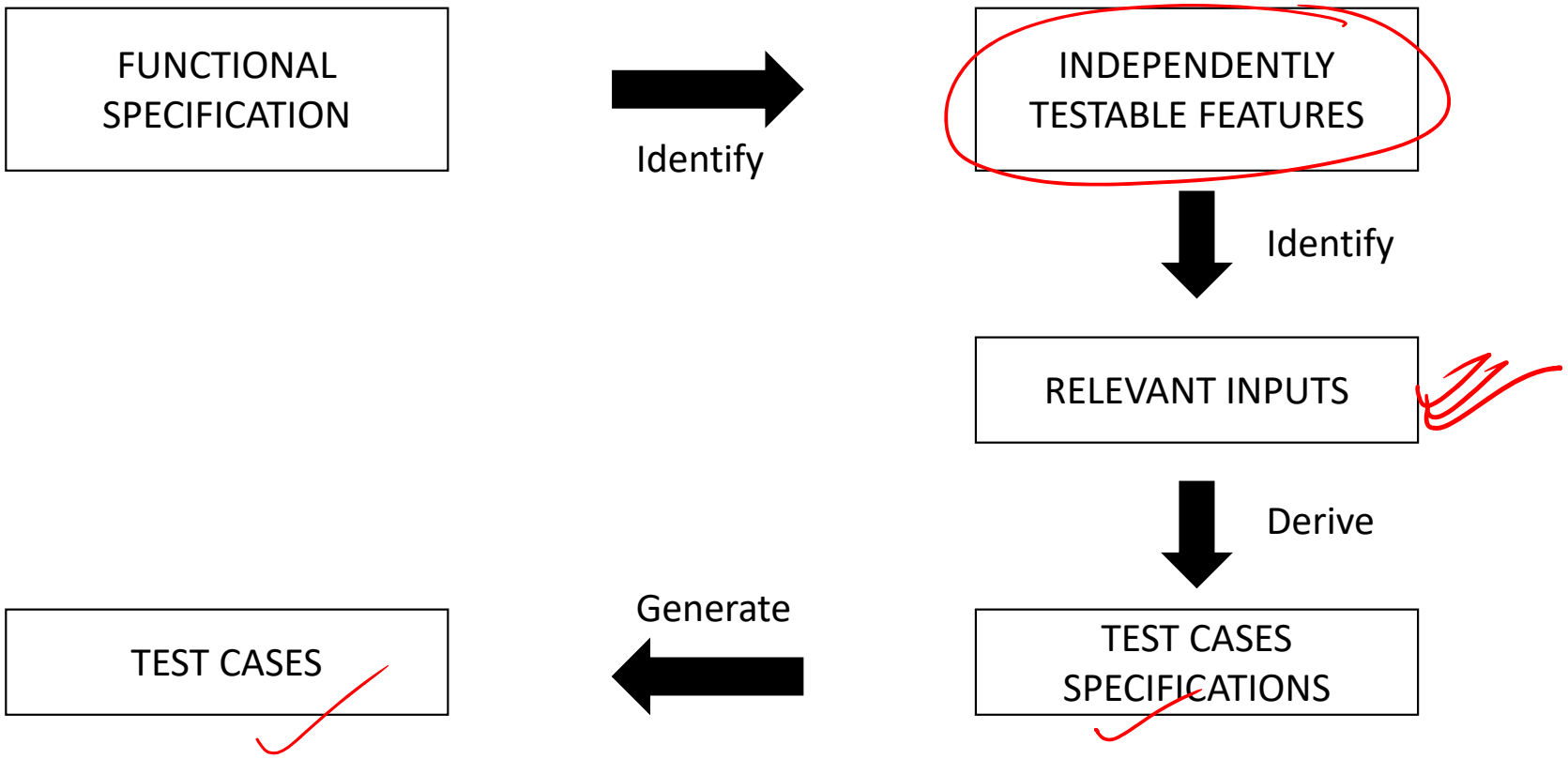
FUNCTIONAL SPECIFICATION



TEST CASES

Slide adapted from Alessandro Orso

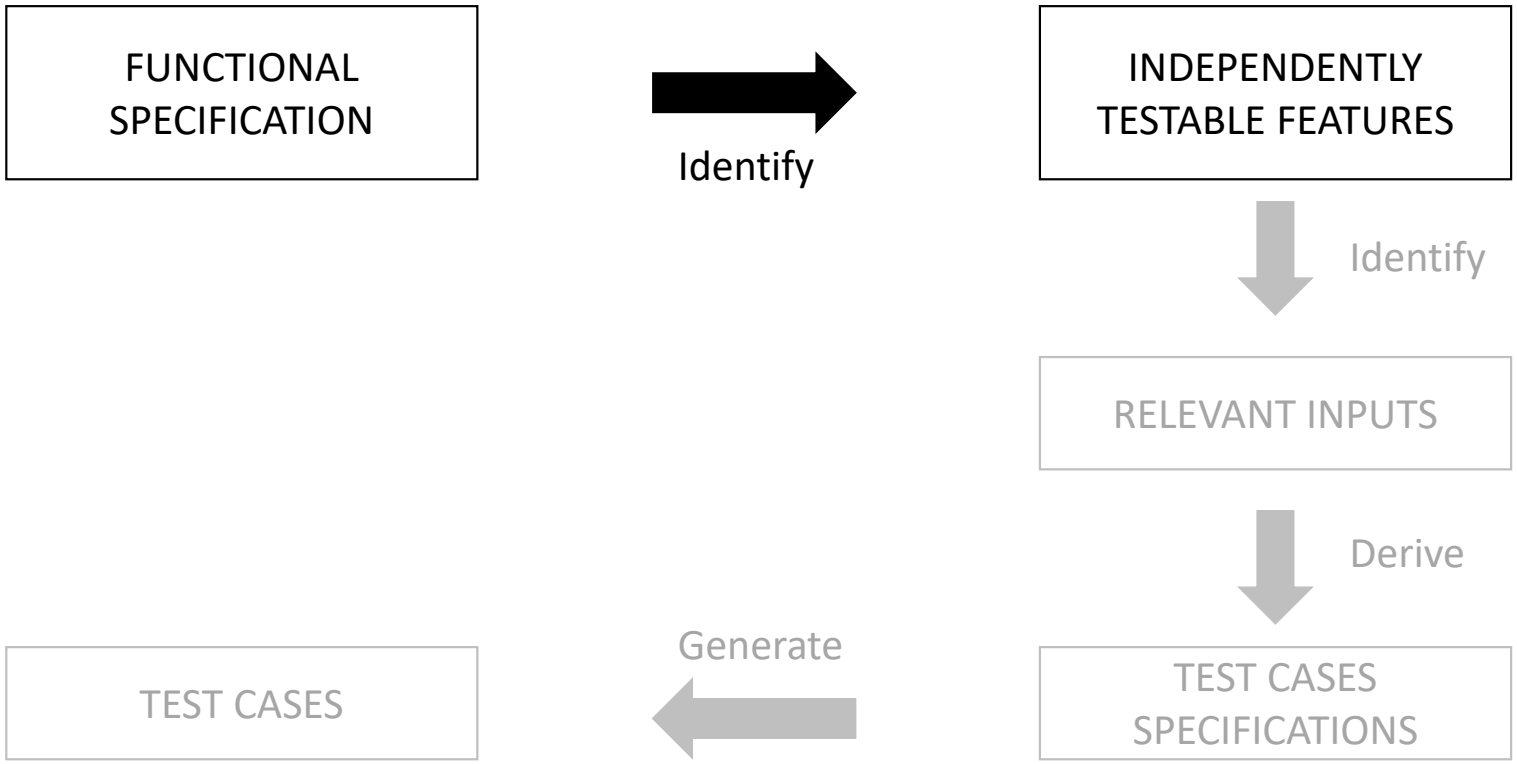
A systematic Functional-Testing Approach



Decoupling; Automated Sub-tasks; Monitor testing process

Slide adapted from Alessandro Orso

A systematic Functional-Testing Approach



Slide adapted from Alessandro Orso

Identifying Testable Features



printSum (int a, int b)

How many independently testable features do we have here?

[✓] 1

[ ] 2

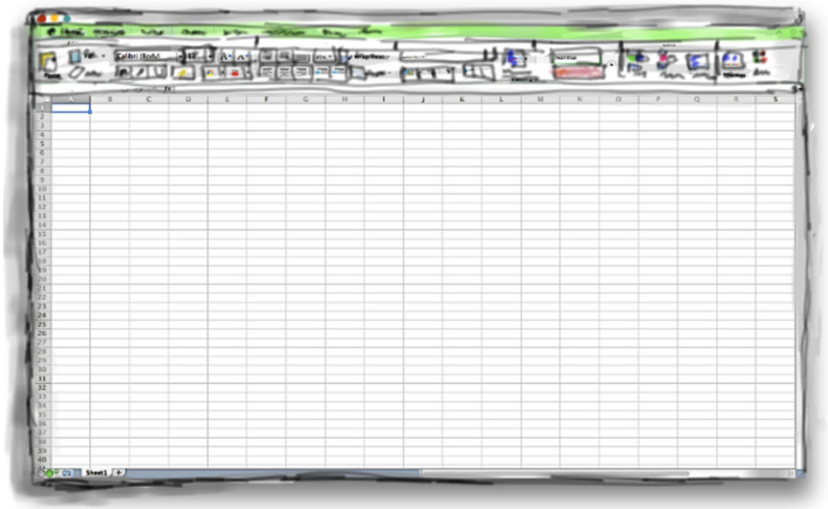
[ ] 3

[ ] 4

Identifying Testable Features



Identify 3 possible independently testable features for a spreadsheet



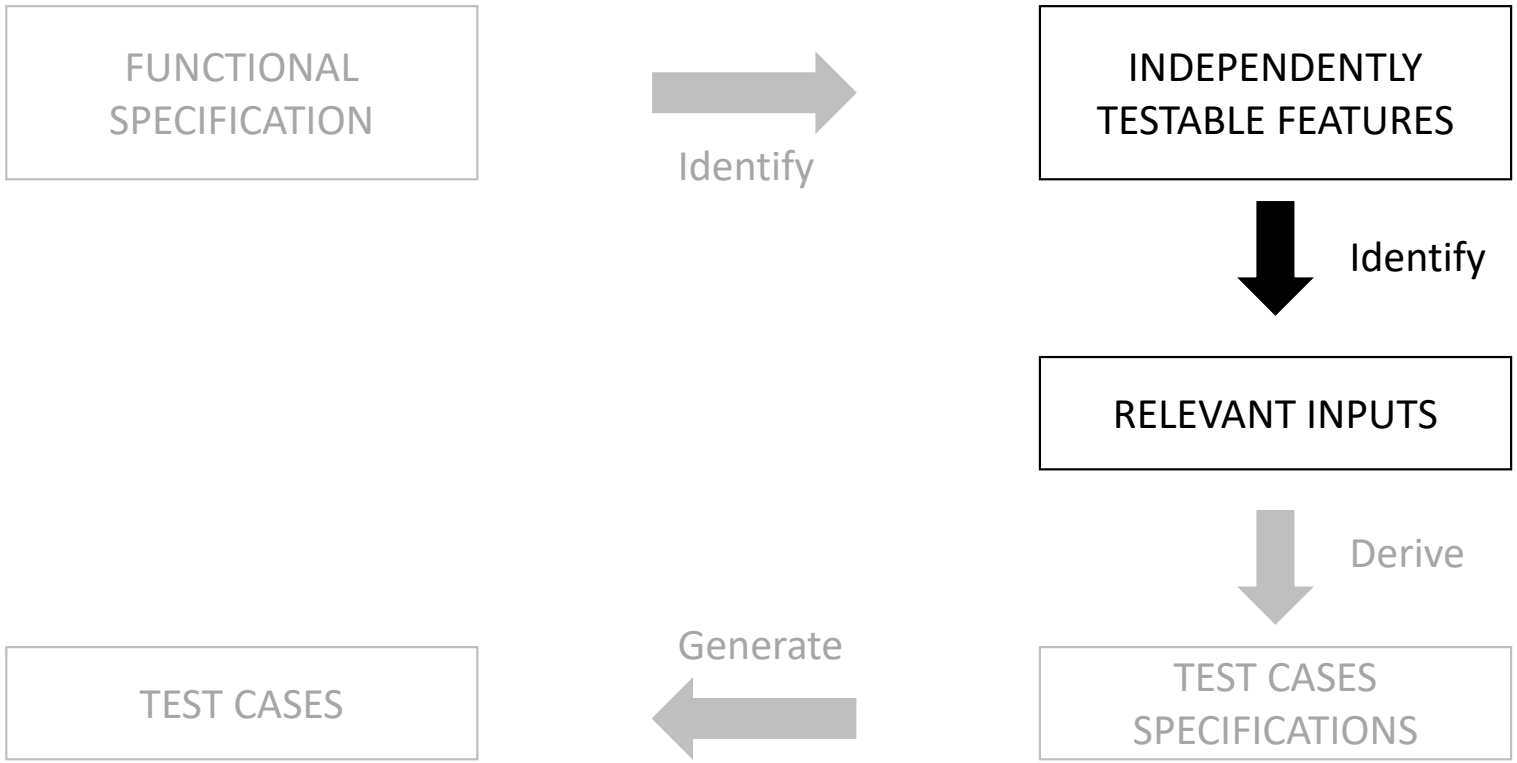
Saving

[Statistical Functions]

[Cell Merging]

[Chart creation]

A systematic Functional-Testing Approach



Slide adapted from Alessandro Orso

Test Data Selection



How to select meaningful set of inputs and corresponding outputs?

Powerful machines, why not exhaustive search?

Straw-Man Idea: Exhaustive Testing!





How long would it take to exhaustively test the function printSum(int a, int b)?

$2^{32} * 2^{32} = 2^{64} \sim= 10^{19}$ tests

1 test per nanosecond

10^9 tests per second

10^{10} seconds overall ~ 600 years

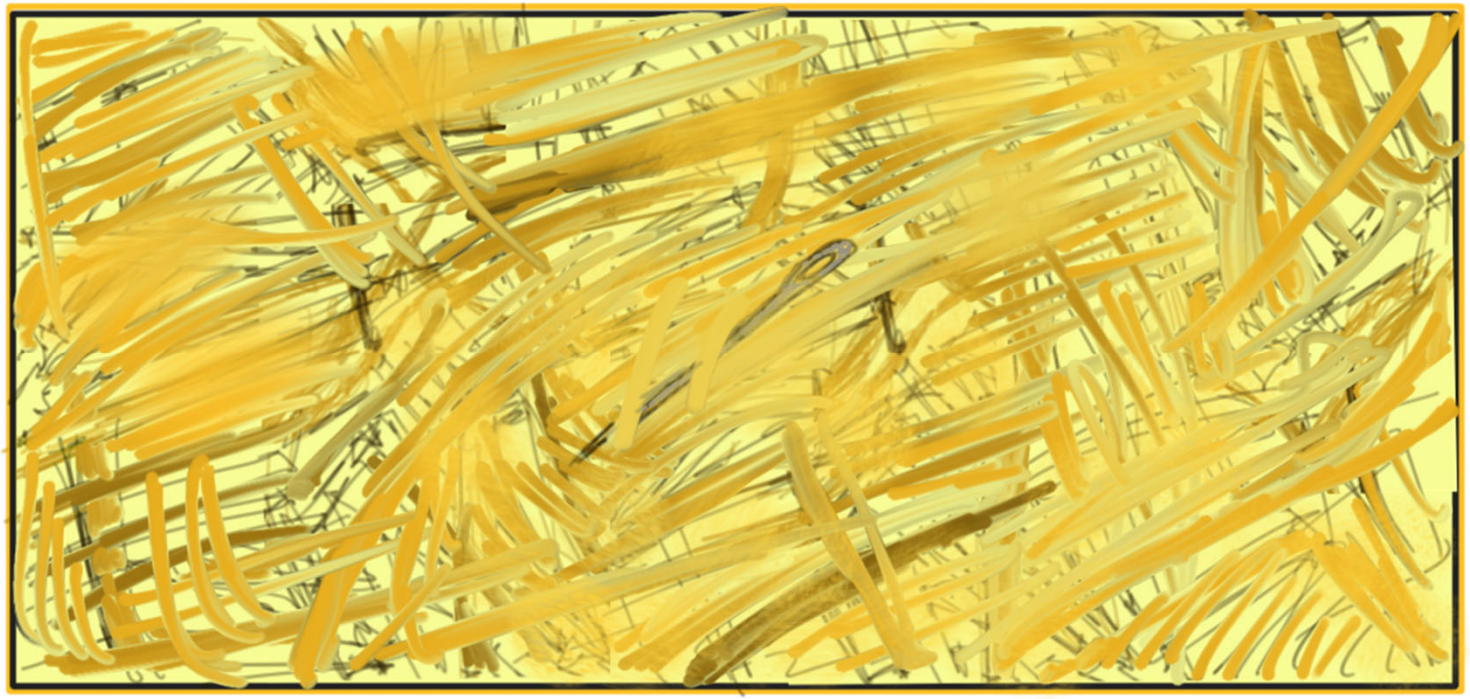
Random Testing



Advantages

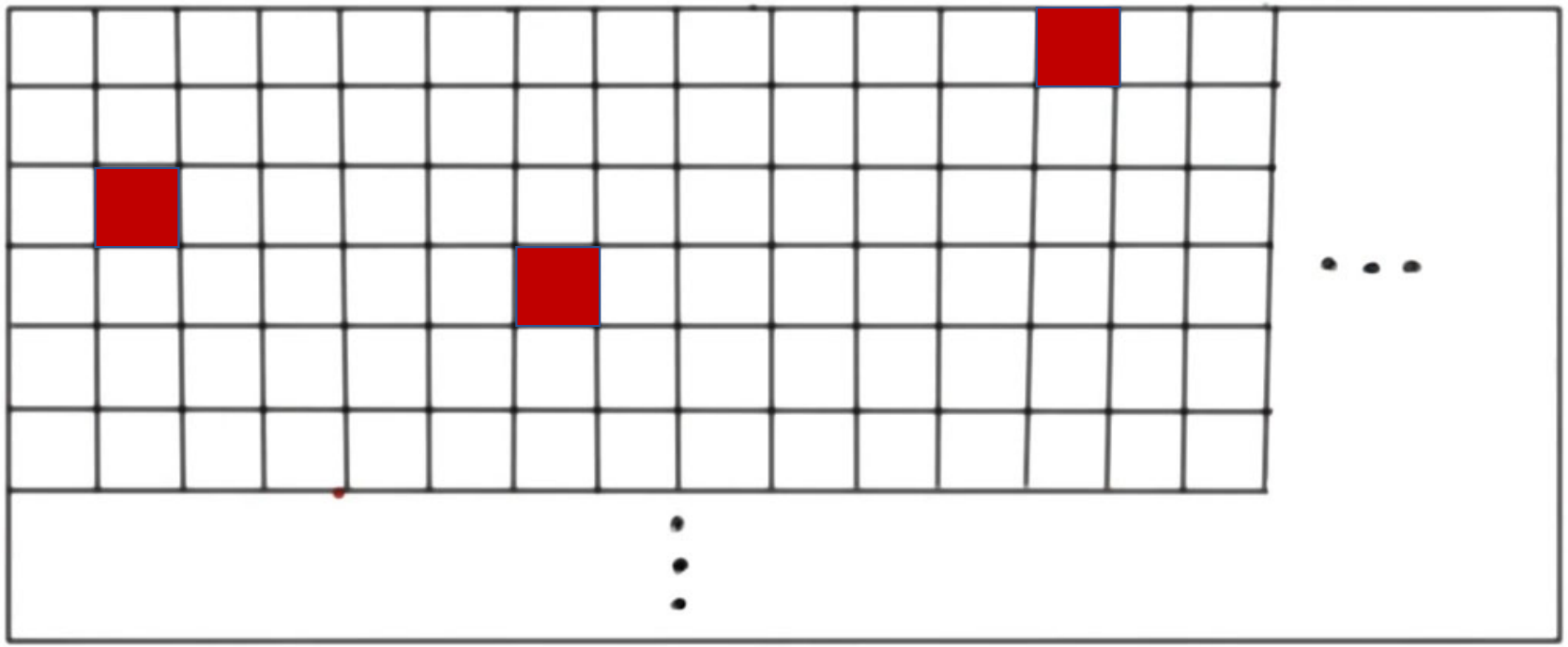
- Pick inputs uniformly
- All inputs considered equal
- No designer bias (developer may develop code based on an assumption, test cases may also be biased)

So why not random?

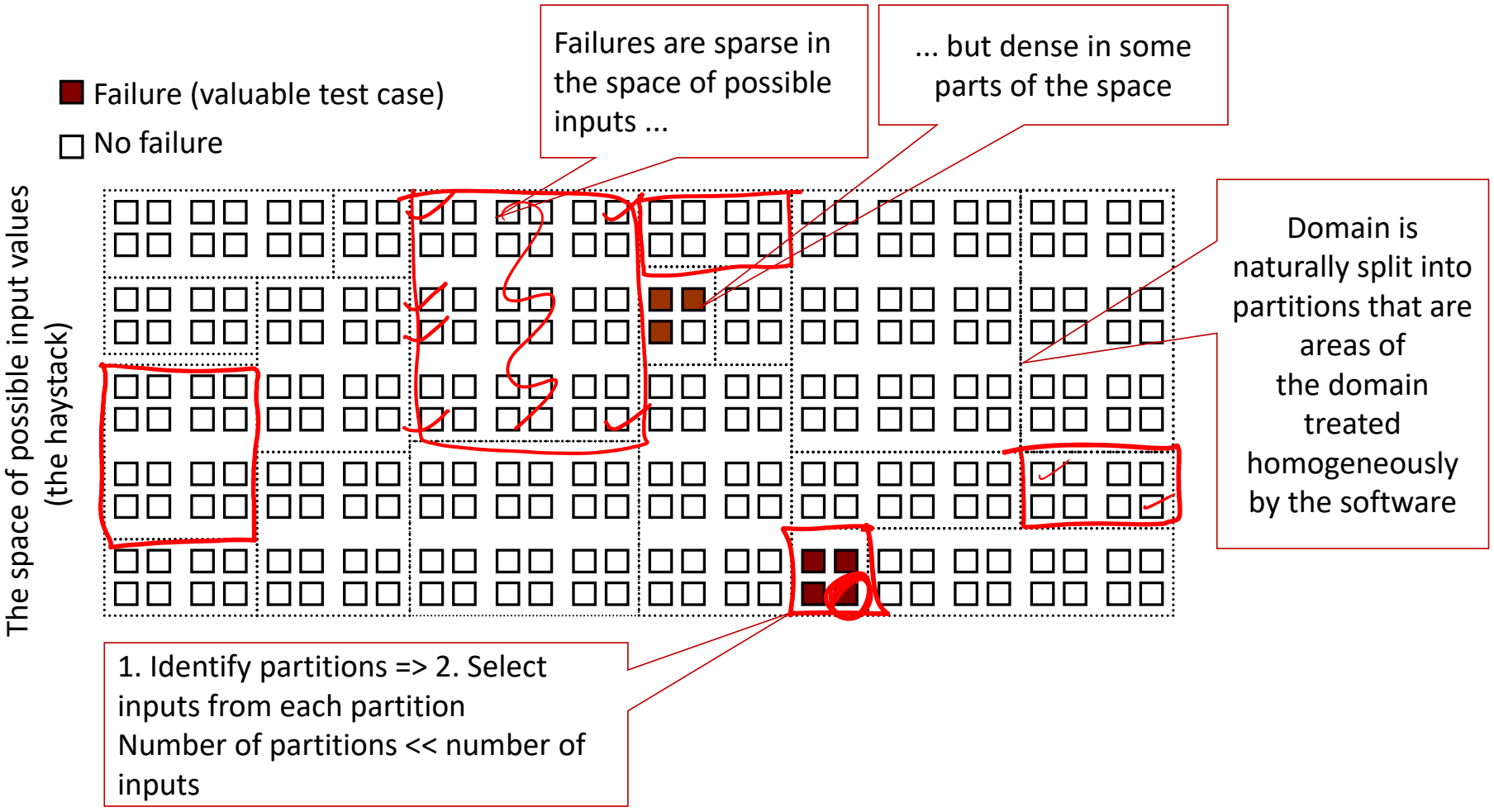


Same as finding many needles in a haystack

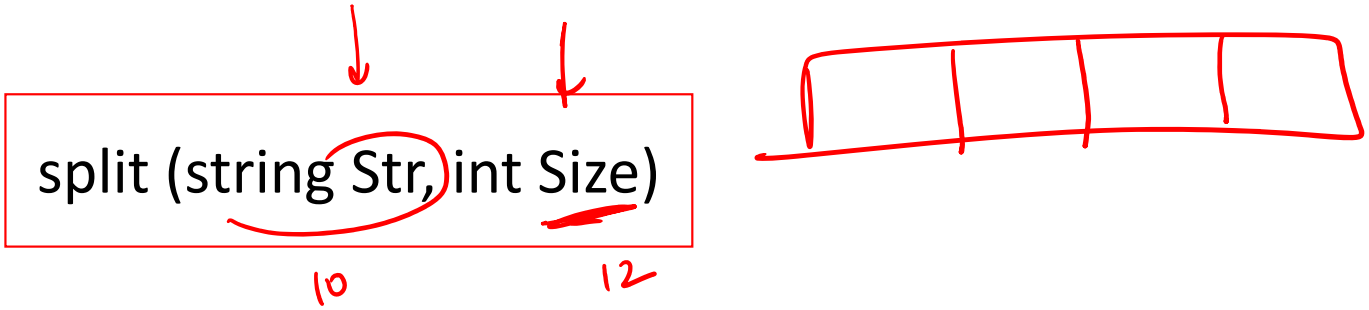
So why not random?



Systematic Partition Testing



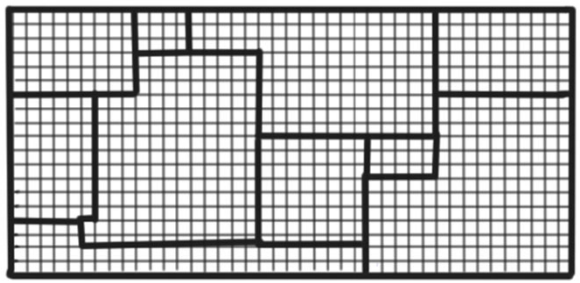
Example



1. Identify partitions:

- Size < 0 (Designer bias might let you not pick this partition)
- Size = 0
- Size > 0
- Str with length 0
- Str with length < Size
- Str with length in [Size,Size*2]
- Str with length > Size*2
- …

Boundary Values



2. Select **interesting** Inputs from each partition

Basic Idea: Errors tend to occur at the boundary of a sub-domain

=> Select inputs at these boundaries

Select interesting Inputs from each partition: **Example**

split (string Str, int Size)

Some possible partitions:

- Size < 0
- Size = 0
- Size > 0

- Str with length < Size
- Str with length in [Size, Size*2]
- Str with length > Size*2

Some possible inputs:

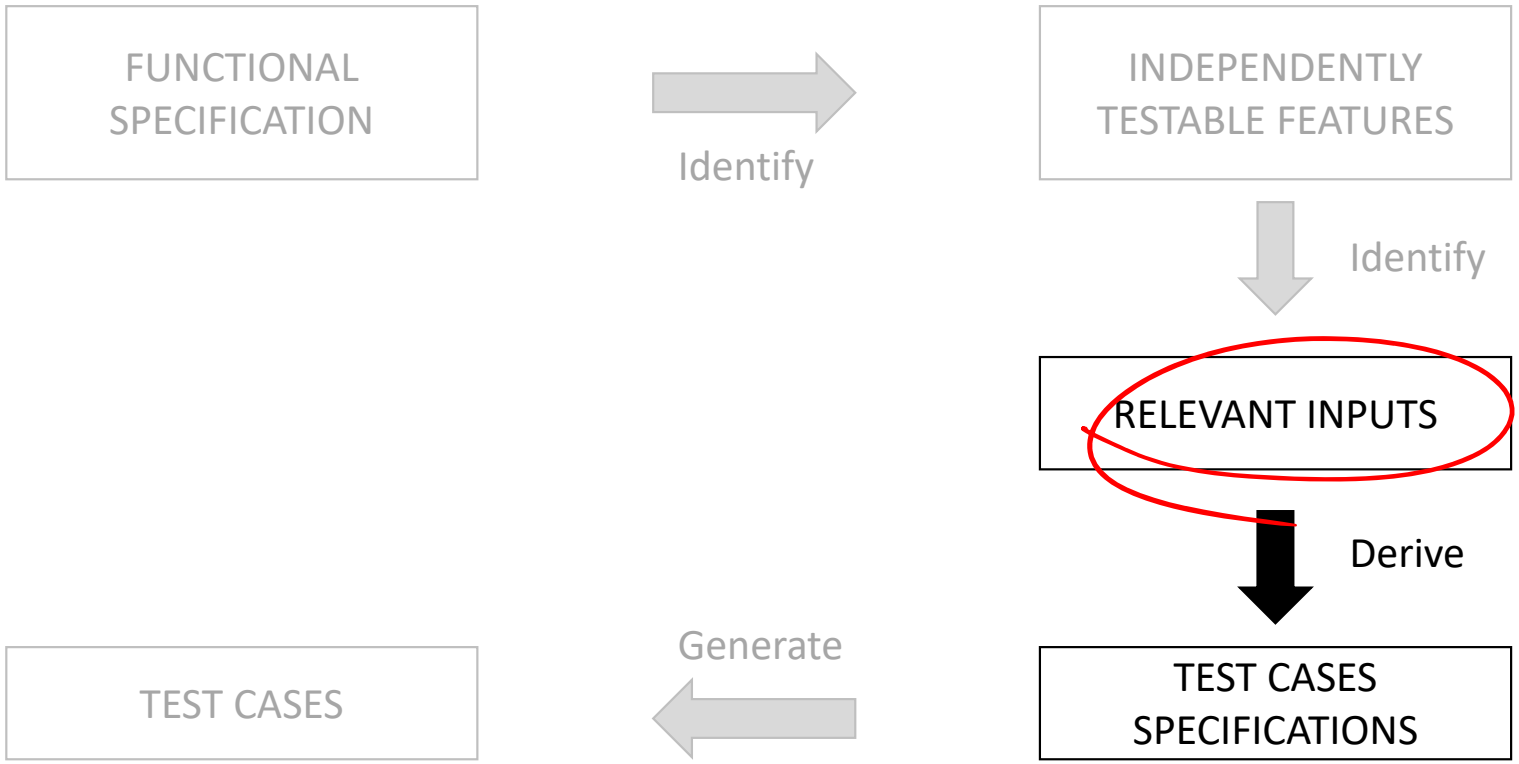
- Size = -1
- Size = 1
- Size = MAXINT

- Str with length = Size- 1
- Str with length = Size
- …

Size x 2

Size x 3

A systematic Functional-Testing Approach



3. Generate Test Case Specifications: Example

split (string Str, int Size)

Some possible inputs:

- Size = -1
- Size = 1
- Size = MAXINT

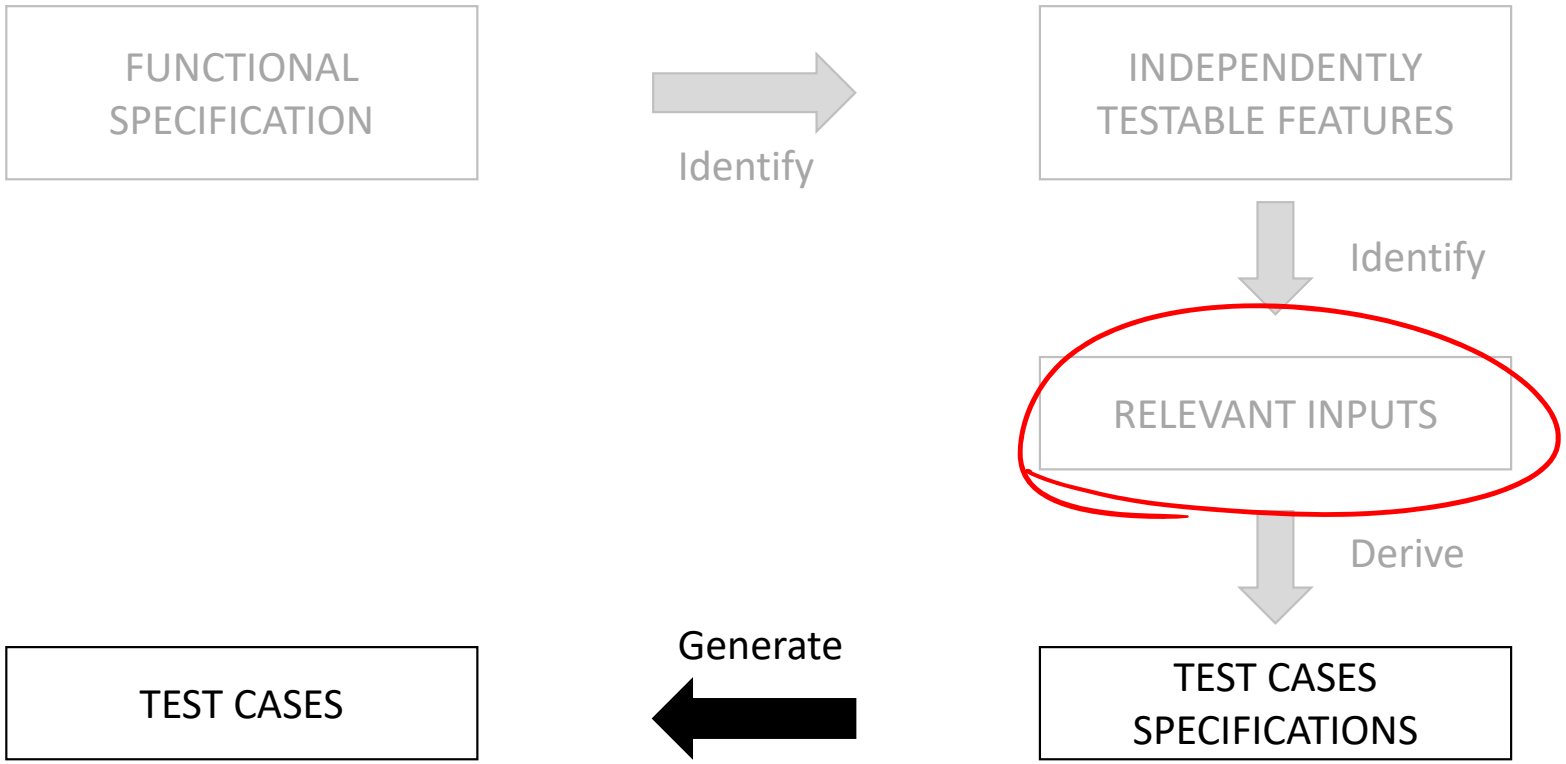
×

- Str with length = Size- 1
- Str with length = Size
- …

Test Case Specifications: (combine input values)

- ~~Size = -1, Str with length = -2~~
- ~~Size = -1, Str with length = -1~~
- Size = 1, Str with length = 0
- Size = 1, Str with length = 1
- …

A systematic Functional-Testing Approach



Slide adapted from Alessandro Orso

A Specific Functional Testing Black-Box Approach
The Category-Partition Method

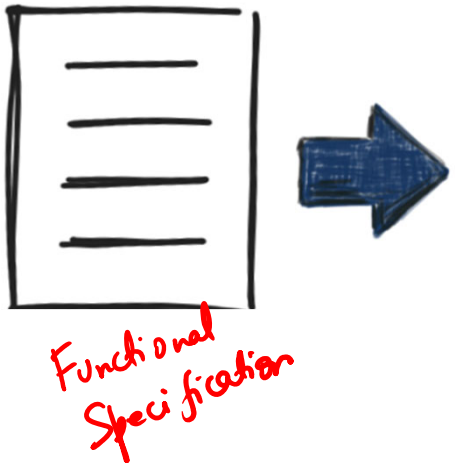
[Ostrand & Balcer, CACM, June 1988]



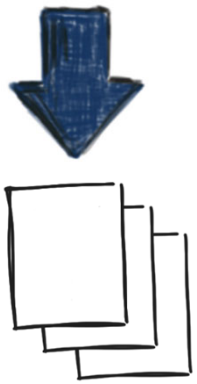
Specification

Test Cases

The Category-Partition Method



1. Identify independently testable features
2. Identify Categories
3. Partition Categories into choices
4. Identify constraints among choices
5. Produce/Evaluate test case specifications
6. Generate test cases from test case specifications



Test Cases

Test case = {i, o}

Identify Categories

Characteristics of each input element

```
split (string Str, int Size)
```

Input Str

- Length
- Content

Input Size

- value

Partition Categories into choices

Interesting cases (subdomains) – boundary values

split (string Str, int Size)

Input Str

- Length
  - 0
  - Size-1
- Content
  - Only Spaces
  - Special characters

Input Size

- Value
  - 0
  - >0
  - <0
  - MAXINT
  - …

Identify Constraints among choices

is to reduce the number of bad/redundant test cases

To Eliminate meaningless combinations & To reduce number of test cases

Three types: PROPERTY---- IF, ERROR, SINGLE

Input Str

- Length
  - 0 — PROPERTY zerovalue
- Content
  - Special characters — If !zerovalue

Input Size

- Value
  - <0 — ERROR
  - MAXINT — SINGLE

look 1

Overall no. of test case with the single constaint should be 1

Produce And Evaluate Test Case Specifications

Can be automated

Produces test frames

Example (specify the characteristic of the inputs for that test)

Test frame #45

Input Str

length: size -1

content: special characters

Input Size

value: >0

Produce and evaluate test case specification

-how many test frames?

-add additional constraints to reduce the number if required

Generate Test Cases from Test Case Specification

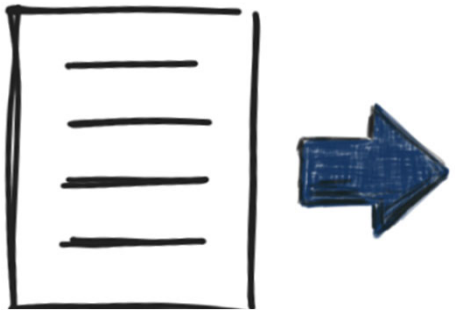
Simple Instantiation of frames

Final result: Set of concrete tests

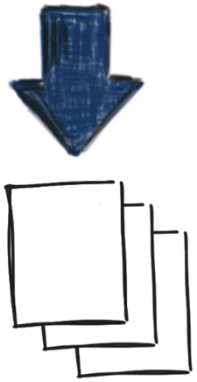
Example (specify the characteristic of the inputs for that test)

Test case #45
Str = “ABCC!\n\t”
Size = 10

The Category-Partition Method



1. Identify independently testable features
2. Identify Categories
3. Partition Categories into choices
4. Identify constraints among choices
5. Produce/Evaluate test case specifications
6. Generate test cases from test case specifications



Test Cases

Category Partition DEMO TIME

- Use category partition to generate test frames from a specification file (with categories, partitions, and constraints)
- Tool called TSLgenerator is used: Developed by team at UC Irvine, Oregon State, and Georgia Tech
- Download from: https://github.com/alexorso/tslgenerator/tree/master/Binaries
- run the code from command prompt: **./TSLgenerator-win8.exe**
- For help: **./TSLgenerator-win8.exe –manpage**
- To get number of test cases and write the test frames against your specification file: **./TSLgenerator-win8.exe -c *filename***

Can AI Generate Good Black-Box Tests?

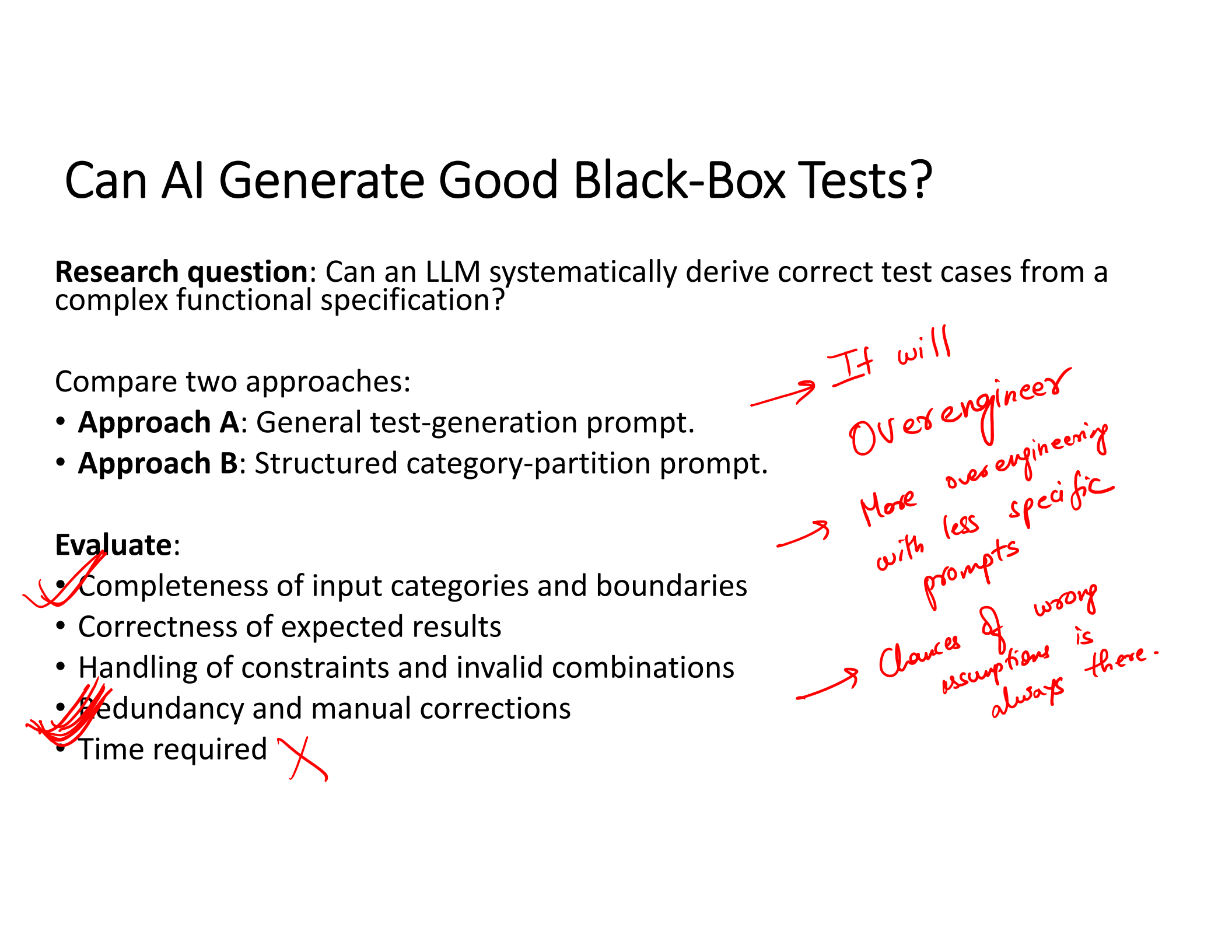
Research question: Can an LLM systematically derive correct test cases from a complex functional specification?

Compare two approaches:

- **Approach A**: General test-generation prompt.
- **Approach B**: Structured category-partition prompt.

Evaluate:

- Completeness of input categories and boundaries
- Correctness of expected results
- Handling of constraints and invalid combinations
- Redundancy and manual corrections
- Time required



Experiment: Discount and Checkout System

A checkout system calculates the final order total using:

- Order subtotal
- Customer type: Regular, Premium, Student
- Promotional code: SAVE10, FLAT20, or absent
- Membership discounts and promotional eligibility
- Discount limits and shipping thresholds
- Input validation and rounding rules

Challenge: Identify meaningful partitions, constraints, and boundary cases.

https://chatgpt.com/share/6ae105f8-f9c0-83ea-b277-392196e652cc

Human Review of AI-Generated Tests

Evaluation	General Prompt	Category-Partition Prompt
Volume	185 scenarios	20 test frames + 35 concrete test cases
Useful test	TC-174: Reordering request fields should not change the result	T-28: $10.05 with `SAVE10` tests half-cent rounding
Additional coverage	TC-173: Repeated calculation should produce the same result	T-08/T-09: Test just below and at a promotional threshold
Limitations	Many overlapping tests; some extend beyond the specified functionality	Introduces assumed discount rates, stacking rules, and shipping fees
Human review needed	Prioritize relevant, nonredundant tests	Verify assumptions, constraints, and expected results

What should a human reviewer catch?

Assumptions

Issue	Example from AI Output
Incorrect test design	General TC-017: Tests whitespace handling without whitespace in the input
Scope expansion	General TC-165: Tests concurrent order requests
Unconfirmed constraints	Category-Partition TF-14: TF-14 proposes combined discounts exceeding the cap without established stacking rules
Invented expected results	Category-Partition T-11: Calculates a $12 total using using illustrative discount and shipping rules

Takeaway: Human review must verify test relevance, feasibility, input correctness, and expected results.

Supplementary - Black-box Testing Example: more effective

User Interface (UI) Testing

- Imagine a simple login form for a web application with the following fields: Username (text field), Password (text field), Login Button
- The form is designed to authenticate users based on their input. The requirements specify that: A username must be between 5 to 15 characters; The password must be at least 8 characters long; Special characters are allowed in the password but not in the username.

Why Black-Box Testing Could Be More Effective:

- **Focus on User Behavior**: Black-box testing is effective here because it focuses on how a user interacts with the form rather than the internal logic behind the input validation.
- **Testing Inputs and Outputs**: It can test various scenarios like valid and invalid usernames, short and long passwords, special characters, and empty fields, verifying that the user gets the correct messages like "Username must be at least 5 characters" or "Password is too short."
- **Real-World Use Cases**: Black-box testing ensures that the form behaves as expected in real-world use cases and catches issues like improper error messages, incorrect form submissions, or failures to handle certain types of input.

Supplementary - Black-box Testing Example: more effective

Why White-Box Testing May Be Less Effective:

- **Focus on Code Paths, Not User Experience**: White-box testing in this scenario would focus on the internal validation logic—checking the functions that verify the username length, password length, and special character restrictions.
- **Potential to Miss UI Issues**: It may miss issues related to the user experience, such as how error messages are displayed or whether the form reloads when it shouldn't.
- **Less Emphasis on Usability**: White-box testing might ensure that the functions for validation work perfectly according to the code, but it could overlook how well the user interface guides the user through input errors or displays feedback.

Supplementary - How can we use AI tools?

1. Using LLM for Test Case Generation

- **Description:** GPT/Llama/Gemini/Claude can generate coherent, contextually relevant text based on prompts. This capability can be utilized to create detailed test cases from a set of functional requirements written in plain English, significantly speeding up the test design process.
- **Example:** For a feature that allows users to book flights, you might have a requirement: “Given --- project description, one of the requirements is “The user should be able to select a departure and return date using a calendar widget.“ Generate test cases for this requirement”. From this, LLM can generate a series of test cases, such as:
  - Test the functionality of the calendar widget for date selection.
  - Check the behavior when no dates are selected.
  - Verify the system's response when past dates are selected.

Supplementary - How can we use AI tools?

2. Selenium with AI Extensions

- **Description:** Selenium is a tool for automating web browsers, allowing it to mimic user actions on a webpage. AI extensions (test.ai, testRigor, Selenium Sage, Parasoft Selenic) can enhance Selenium by predicting UI changes and optimizing test flows based on previous test runs, reducing test maintenance.
- **Example:** For a web application's login page:
- Typical Selenium Test:
  - Enter valid credentials and verify successful login.
  - Enter invalid credentials and check for error messages.
- With AI Extensions:
  - The AI identifies frequent UI changes, like button relocations or text field adjustments, and dynamically updates the Selenium selectors before running tests.
  - Predicts potential fail points like heavy load times for login button clicks during peak hours and adjusts test parameters dynamically.

Supplementary - How can we use AI tools?

3. Postman and Postbot for API Testing

- **Description:** Postman is a popular tool for API testing that allows users to construct complex HTTP requests, examine the responses, and automate testing through scripts. Postbot, Postman's AI assistant, can generate test scripts based on the user's API schema or past test scripts.
- **Example**: For an API endpoint that retrieves user profiles:
  - API Endpoint: GET /api/user/{userID}
  - Typical Test Cases Using Postman:
    - "Validate response with a valid userID."
    - "Check response for a userID that does not exist."
  - AI-Generated Test Cases by Postbot:
    - "Test for SQL injection vulnerabilities by inputting SQL code as the userID."
    - "Verify the handling of unusually long user ID strings."